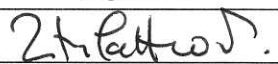
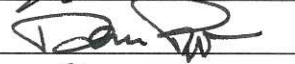


TITLE: GXP and Non-GXP Study Report Review and Approval Form

DOCUMENT NO.	VERSION	ISSUE DATE	EFFECTIVE DATE	PAGE
GE003FM01	02	17 Oct 2014	20 Oct 2014	2 of 2

☐ Vendor Study Protocol		☒ Vendor Study Report		☐ Lpath Study Report	
Lpath Study No.: LT3114-PHA-011-R		Version No.: 01		☒ Non GXP ☐ GCP	☐ GLP ☐ GMP
Product Name: LT3114		Test Article: 504B3 (B3) ; LT1015			
Study Title: Dose response efficacy of test compound in a therapeutic model of diabetes-induced neuropathic pain					
Study Vendor/Site: ☐ N/A Washington Biotech		Study Director: ☐ N/A Rosalia Matteo		Study Contributor(s): ☒ N/A	
Vendor Study No./Version: ☐ N/A DIP-LPI-1		Study Dates: Nov 2012-Feb 2013		Report Date: March 2013	
Lab Notebook Reference: LNB# 248 pp. 79-82					☐ N/A
Executive summary of study results: ☐ See Attached ☐ N/A					
Intravenous injection with 55mg/kg streptozocin resulted in diabetes in rats seven days later. Approximately one month after disease induction the diabetic rats exhibited peripheral neuropathy as indicated by increased sensitivity to thermal and mechanical stimulation. Twice weekly intravenous injection with vehicle or antibodies was initiated five weeks after disease initiation and continued for four weeks. Twice weekly injection with 30 mg/kg LT1015 (nonspecific IgG2b control Ab) had no ameliorative effect on the diabetic peripheral neuropathy (DPN). On Day 50 rats in this group were injected subcutaneously with 30mg/kg gabapentin thirty minutes prior to evaluation of the peripheral response to thermal and mechanical stimulation. This treatment resulted in a significant increase in the paw pressure required to elicit a response to mechanical (von Frey) stimulation but had no effect on the response to thermal stimulation (Hargreaves test), relative to the saline-treated animals. Twice weekly intravenous injection with 504B3 (IgG2b murine anti-LPA antibody), starting five weeks after disease induction, had a transient effect on the peripheral neuropathic pain associated with STZ-induced diabetes at the highest dose (30mg/kg). Twice weekly intravenous injection with 30mg/kg LT3114 (human anti-LPA antibody), starting five weeks after disease induction, was efficacious in reducing the sensitivity of diabetic rats to both thermal and mechanical stimulation.					

LPATH APPROVAL

Function	Name (Print)	Signature	Date
Study Director/Monitor	Rosalia Matteo		11-26-2014
Contributor	N/A		
Research	GARY WOODWORTH		26 th November 2014
Development	DARIO PAGLIARINO		12/1/2014
Quality Assurance	ELVIRA MARQUEZ		01 DEC 14



WASHINGTON BIOTECHNOLOGY, INC.

Mailing Address:

P.O. Box 211
Simpsonville, MD 21150

Laboratories & Shipping Address:

6200 Seaforth Street.
Baltimore, MD 21224
Phone: 410.633.9210
Fax: 410.633.9213
E-mail: vlowe@washingtonbiotech.com

**FINAL REPORT: DOSE-RESPONSE EFFICACY OF TEST COMPOUND IN A
THERAPEUTIC MODEL OF DIABETES-INDUCED NEUROPATHIC PAIN**

Project Title: DIP-LPI-1

Project: To evaluate dose-response therapeutic efficacy of Lpath, Inc. test compounds in a rat model of diabetes-induced peripheral neuropathic pain. This study was carried out in male Sprague-Dawley rats according to Washington Biotechnology, Inc. Protocol 1090.

Sponsor: Rosalia Matteo, Ph.D.
Senior Scientist
Lpath, Inc.
4025 Sorrento Valley Blvd
San Diego, CA 91913

Summary of Results: Twice weekly intravenous injection with 30 mg/kg LT1015 (nonspecific IgG2b Control), starting five weeks after disease induction, had no effect on the peripheral neuropathic pain associated with Streptozocin-induced diabetes.

Twice weekly intravenous injection with 504B3 (IgG2b murine anti-LPA antibody), starting five weeks after disease induction, had a transient effect on the peripheral neuropathic pain associated with Streptozocin-induced diabetes at the highest dose (30 mg/kg).

Twice weekly intravenous injection with 30 mg/kg LT3114 (Human anti-LPA antibody), starting five weeks after disease induction, was efficacious in reducing the sensitivity of diabetic rats to both thermal and mechanical stimulation.

Summary of Procedures:

1. November 15, 2012: Received via Federal Express on coolants from Lpath, Inc. and stored at 4-8°C:

 - 24 cryovials LT1015, lot HR121332, 12.74 mg/ml
 - 34 cryovials 504B3, lot HR111266, 11.98 mg/ml
 - 34 cryovials LT3114, lot LP101202, 9.43 mg/ml
2. November 20, 2012: 70 Sprague-Dawley rats (Harlan Sprague-Dawley, Inc., PO #659023, R #3185, 170-190g, male) were received, individually examined, and housed in fourteen cages of five rats each. Each animal was in apparent good health: no signs of clinical disease or distress. The animals were placed in quarantine under 12 hr light/dark conditions with daily inspections, provided with *ad libitum* access to food and water.
3. November 26, 2012: The animals were individually examined and observed to be free of clinical signs of disease or distress. The animals were released to routine maintenance. No deaths were recorded during the quarantine period.
4. November 27, 2012: The rats were ear notched (SOP 800) for identification purposes and weighed. Eight animals (rats #1-7, 11) were assigned to Group 1 (non-diseased) based upon average weight.
5. November 28, 2012: The body weight and baseline response to thermal stimulation using the Hargreaves apparatus (IITC, model 336, SN 70108) was recorded for all rats (RESULTS). The rats were acclimated (15 minutes) to the apparatus which consisted of six separated Plexiglas chambers on a glass table. A mobile radiant heat source located under the table was focused under the desired hind paw. Paw withdrawal latency (seconds) was recorded three to five times for each hind paw and the average taken as the value for that limb. A timer automatically activated with the light source and the latency was defined as the time required for an abrupt withdrawal of the hind limb. The light source was calibrated to yield a normal latency of approximately 10-12 seconds; to prevent tissue damage, the light source automatically cut off at 20 seconds.

6. November 29, 2012: The body weight and baseline response to mechanical stimulation using von Frey fibers was recorded for all rats (RESULTS). Each rat was placed in a chamber with a metal wire mesh floor and was acclimated for 15 minutes. Starting with a probe diameter of 1.65 (0.008 grams of pressure) to a diameter of 6.65 mm (300 grams of pressure), the probes were systematically applied to the plantar surface of the hind paw until paw withdrawal occurred.
7. December 4-6, 2012: The rat weights were recorded daily.
8. December 6, 2012: 0.01M citrate buffer was prepared by dissolving 0.384 g citric acid (Fisher Scientific, Cat. A940-500, lot 025173) in 199.86 ml de-ionized water. The pH was adjusted to 4.5 with 3N NaOH.
9. December 7, 2012: 1.22 g streptozocin (STZ, Sigma Chemical Co., Cat. S0130, lot SLBB7526V) was dissolved in 110.909 ml 0.01M citrate buffer to prepare a 11 mg/ml solution which was stored at 4-8°C when not in use.
- DAY 0 The rat weights were recorded (RESULTS). The rats in Group1 were injected intravenously (IV, SOP 1650) with 5 ml/kg 0.01M citrate buffer. The remaining rats were injected IV with 5 ml/kg of STZ solution (55 mg/kg). All rats were individually housed on elevated flooring with daily cage changes.
10. December 11, 2012: Rat #10 was found dead.
11. December 14, 2012: The rats were anesthetized (SOP 1810), bled (SOP 1660) via retro-orbital sinus, and blood glucose recorded (RESULTS, One Touch Ultramini, lot 3364714, exp. 0482014). All STZ-injected rats had blood glucose values in excess of 300 mg/dl and were defined as diabetic per the parameters of the protocol.
- DAY 7
12. December 24, 2012: Rat #39 was found dead.
13. January 2, 2013: The response to thermal stimulation using the Hargreaves apparatus was recorded for all rats (RESULTS).
- DAY 26
14. January 3, 2013: The response to mechanical stimulation using von Frey fibers was recorded for all rats (RESULTS). Since the diabetic rats were allodynic, dosing was not initiated.
- DAY 27

15. January 7, 2013:
DAY 31 The response to thermal stimulation using the Hargreaves apparatus and mechanical stimulation using von Frey fibers was recorded for all rats (RESULTS).

16. January 8, 2013: 6 vials LT1015 was diluted into 6.74 ml sterile PBS (Sigma, Cat. D8537, lot RNBC3873, exp 05/2014) to prepare a 6 mg/ml solution for Group 3.

2 vials 504B3 was diluted into 9.98 ml sterile PBS to prepare a 2 mg/ml solution for Group 4.

6 vials 504B#3 was diluted in 6 ml sterile PBS to prepare a 6 mg/ml solution for Group 5.

8 vials LT3114 was diluted into 4.573 ml sterile PBS to prepare a 6 mg/ml solution for Group 6.

DAY 32 The rats were sorted into treatment groups based upon the left paw Hargreaves response and body weights recorded (RESULTS). The rats in Groups 1, 2 were injected intravenously (IV, SOP 1650) with 5 ml/kg sterile PBS. The remaining rats were injected IV with 5 ml/kg of their respective antibody.

TABLE 1: TREATMENT GROUPS

Group	Diabetic	Treatment	Dose (mg/kg)
1	No	PBS	5 ml/kg
2	Yes	PBS	"
3	"	LT1015	30
4	"	504B3	10
5	"	"	30
6	"	LT3114	30

17. January 11, 2013:
DAY 35 Test materials were prepared as previously mentioned. The rats were weighed (RESULTS) and injected IV with 5 ml/kg of their respective preparation.

18. Jan. 14-Feb. 1, 2013:
DAYS 38-56 The rats were evaluated for their responses to thermal and mechanical stimulation every Monday (RESULTS). After the last measurement was recorded, the rats were anesthetized (SOP 1810) and bled (SOP 1661) for sample into pre-chilled EDTA microtainer tubes (Becton Dickinson, Cat. 365973, lot 2201348, exp. 12/2013). The blood was processed to

plasma (SOP 6200) which was stored in labeled Eppendorf tubes at -80°C (SOP 7020).

Fresh dosing solutions were prepared every Tuesday and Friday, the rat weights were recorded, and the rats were injected IV with 5 ml/kg of their respective preparation as in TABLE 1.

19. January 16, 2013: Received via Federal Express on dry ice from Lpath, Inc. and stored at -80°C:

- 48 cryovials LT1015, lot HR121332, 12.74 mg/ml
- 59 cryovials 504B3, lot HR111266, 11.98 mg/ml
- 36 cryovials LT3114, lot LP101223, 14.2 mg/ml

20. January 21, 2013: Rat #34 in Group 2 was found dead.
DAY 45

21. January 29, 2013: Rat #31 in Group 3 was found dead.
DAY 52

22. February 4, 2013: 100 mg gabapentin (Sigma, Cat. G154, lot 128K1528V) was dissolved in 10 ml de-ionized water to prepare a 10 mg/ml solution.

DAY 59
The rats were evaluated for their responses to thermal and mechanical stimulation (RESULTS). The rats in Group 3 were evaluated 30 minutes after a subcutaneous (SOP 1610) injection with 3 ml/kg (30 mg/kg) gabapentin.

23. February 5, 2013: The rats were anesthetized (SOP 1810) and exsanguinated (SOP 1686) into pre-chilled EDTA tubes (Becton Dickinson, Cat. 366643, lot 2247274, exp 08/2013). The blood was processed to plasma (SOP 6200) which was stored in labeled 15 ml conical tubes at -80°C (SOP 7020). Segments (~3-5 mm) of the left and right sciatic nerve were removed to individually labeled Eppendorf tubes, frozen, and stored at -80°C. The spinal cords were flushed from the spinal column, frozen, and stored in labeled 15 ml conical tubes at -80°C.

24. February 11, 2013: The frozen plasma, nerve tissue, spinal cords, and remaining test materials were shipped on dry ice to Lpath, Inc. via overnight Federal Express courier service. Samples are stored at Lpath for further analysis.

Results:1. Effect of Disease and Treatment on Rat Weight (g):

Group	Rat	STUDY DAY														
		-10	-9	-8	-3	-2	-1	0	32	35	39	42	46	49	53	56
1	1	231	249	256	293	291	298	304	430	430	461	476	447	463	476	477
	2	252	268	277	322	324	330	344	456	461	433	444	481	495	504	510
	3	243	258	260	298	294	302	312	378	378	376	380	378	389	396	407
	4	256	292	280	326	326	341	351	526	534	526	533	535	552	565	571
	5	254	269	275	315	312	318	358	436	430	409	409	407	420	428	435
	6	266	282	288	327	330	336	347	515	516	511	494	474	473	474	492
	7	253	264	268	293	281	292	299	385	386	388	392	394	403	419	430
	11	246	260	266	296	292	300	307	382	395	406	420	434	444	463	472
2	14	255	269	277	325	328	332	344	336	339	326	323	319	328	312	311
	15	256	269	275	314	305	315	323	292	295	276	290	278	284	290	294
	33	236	246	253	296	304	310	314	361	348	339	351	331	343	345	343
	34	244	261	269	306	303	310	319	342	345	328	343	fd			
	41	231	245	252	291	292	298	305	260	257	247	248	248	250	254	245
	43	267	228	257	329	331	338	343	321	330	304	310	289	307	294	298
	48	261	270	276	319	318	325	331	311	308	308	313	297	300	295	290
	69	253	270	277	320	318	325	336	359	338	325	334	325	329	329	344
3	22	253	269	270	306	305	312	323	300	296	293	287	286	294	288	277
	27	254	267	273	316	316	311	330	310	307	285	292	280	280	277	264
	29	258	269	274	313	315	319	332	297	294	287	285	276	275	259	263
	31	233	255	263	298	297	300	311	296	285	275	284	262	250	fd	
	36	236	266	273	299	300	326	314	280	278	263	271	269	263	259	267
	46	249	261	270	320	322	328	338	319	313	288	286	280	268	272	262
	47	251	262	271	328	337	342	350	332	323	328	305	292	280	282	287
	64	258	272	282	333	333	340	351	335	313	319	319	315	307	308	302

fd = found dead

1. Effect of Disease and Treatment on Rat Weight (g): *continued*

Group	Rat	STUDY DAY														
		-10	-9	-8	-3	-2	-1	0	32	35	39	42	46	49	53	56
4	19	239	253	257	307	302	314	326	321	318	310	272	293	293	299	285
	35	251	278	285	332	335	341	351	312	314	306	310	291	291	293	289
	37	264	276	283	333	333	340	350	318	316	308	316	306	303	313	303
	54	246	260	263	307	307	313	326	340	333	329	323	330	319	310	325
	58	246	257	261	311	313	320	330	308	297	286	280	270	277	260	259
	59	267	280	289	345	348	355	360	372	372	338	358	330	346	344	356
	60	240	249	259	306	306	313	322	280	271	270	278	263	279	265	256
	68	244	255	263	316	286	291	297	262	254	246	239	240	236	234	235
5	8	261	260	270	315	310	319	330	289	295	286	290	265	260	257	260
	16	250	260	270	308	306	310	321	242	244	233	235	215	212	210	211
	20	240	253	256	299	296	305	318	238	236	229	230	223	227	225	223
	23	261	274	280	323	322	326	333	324	322	316	320	316	303	310	304
	24	252	264	273	322	323	335	345	327	292	289	288	270	278	268	264
	32	251	261	270	315	321	326	338	312	308	305	290	290	285	281	271
	40	245	263	261	301	301	306	314	267	258	267	254	252	249	250	241
	57	237	244	253	301	300	305	317	282	278	270	273	266	262	254	262
6	13	240	256	262	318	315	324	326	322	312	300	309	288	301	272	270
	28	259	277	283	343	347	328	369	345	345	344	349	336	345	345	332
	42	235	244	253	299	299	308	314	316	318	300	302	290	292	293	289
	44	235	248	255	302	300	309	322	290	288	284	278	284	296	293	292
	45	243	250	254	281	282	289	296	232	228	223	220	207	242	185	182
	55	239	242	252	309	308	315	327	324	278	256	260	268	207	270	260
	61	253	260	270	316	318	326	333	301	296	299	279	280	266	263	252
	67	257	264	278	287	310	317	323	280	274	254	256	245	285	236	236

2. Effect of Streptozocin Injection on Blood Glucose Levels (mg/dl):

Rat	Glucose	Rat	Glucose	Rat	Glucose	Rat	Glucose	Rat	Glucose
1	101	15	404	29	416	43	503	57	406
2	134	16	521	30	477	44	453	58	484
3	137	17	>600	31	467	45	508	59	421
4	114	18	521	32	442	46	480	60	466
5	132	19	461	33	541	47	426	61	419
6	132	20	464	34	449	48	493	62	443
7	110	21	466	35	404	49	374	63	542
11	103	22	>600	36	451	50	464	64	445
8	495	23	523	37	420	51	538	65	468
9	436	24	582	38	478	52	541	66	580
10	fd 12/11	25	499	39	406	53	388	67	415
12	548	26	449	40	454	54	521	68	589
13	487	27	470	41	502	55	480	69	487
14	412	28	490	42	461	56	447	70	553

3. Effect of Disease and Treatment on Left Hind Paw Response to Thermal Stimulation (latency; in seconds):

Group	Rat	STUDY DAY						
		0	28	31	38	45	52	59
1	1	13.21	10.44	10.93	10.95	10.75	13.20	17.26
	2	8.60	10.58	12.27	12.29	12.04	14.34	14.46
	3	10.05	12.63	13.61	13.77	13.12	10.95	14.24
	4	8.01	10.71	11.06	11.77	10.14	15.27	18.46
	5	12.40	10.48	11.47	11.08	11.63	14.84	15.92
	6	8.99	11.96	12.57	19.71	15.22	17.70	19.09
	7	9.55	9.48	10.47	12.16	10.54	13.22	11.16
	11	8.44	12.24	8.22	11.80	9.94	11.55	10.65
2	14	10.63	12.51	8.29	6.40	8.55	8.93	10.69
	15	9.77	10.91	7.37	7.61	7.80	8.30	9.20
	33	7.27	15.18	6.64	5.16	8.30	8.36	9.37
	34	10.35	10.75	5.26	6.22	8.71		
	41	7.00	10.71	7.37	8.79	8.79	9.82	8.73
	43	9.45	10.40	9.84	4.25	6.41	11.58	10.18
	48	10.41	15.64	8.32	8.08	5.36	8.59	9.62
	69	5.47	7.78	6.73	8.25	8.01	7.85	12.18

3. Effect of Disease and Treatment on Left Hind Paw Response to Thermal Stimulation
(latency; in seconds):

Group	Rat	STUDY DAY						
		0	28	31	38	45	52	59
3	22	6.27	13.69	6.78	8.12	6.99	6.50	10.09
	27	8.17	11.40	7.38	10.23	8.51	5.94	12.39
	29	6.23	14.15	9.67	10.68	9.87	14.05	8.81
	31	9.11	14.32	5.36	6.07	5.74	19.17	
	36	8.55	10.79	8.15	7.46	8.54	15.06	16.44
	46	5.25	11.14	6.56	5.67	6.89	7.07	11.93
	47	7.51	16.39	8.42	6.83	8.36	7.14	7.55
	64	8.29	12.45	7.30	7.86	9.28	7.87	8.10
4	19	8.75	10.71	7.55	10.69	7.84	8.73	12.50
	35	9.92	11.44	7.07	10.43	5.95	9.44	7.13
	37	5.83	13.64	7.93	9.20	8.88	7.49	11.40
	54	7.24	11.18	9.61	6.48	7.00	9.38	11.28
	58	7.48	13.79	5.43	9.45	7.62	7.03	11.07
	59	8.40	11.14	6.56	8.59	6.39	9.43	9.04
	60	9.07	12.36	7.25	8.13	7.34	7.96	8.02
	68	8.30	9.14	8.58	6.33	8.02	8.70	10.06
5	8	6.17	12.01	5.60	10.64	8.25	7.83	9.65
	16	10.60	12.99	7.78	10.20	9.74	18.71	12.66
	20	13.16	11.52	9.42	11.30	10.57	10.51	10.57
	23	8.00	12.22	7.57	11.62	8.87	10.66	7.59
	24	6.00	11.33	7.21	8.51	10.86	12.86	10.73
	32	5.22	16.72	7.12	8.12	9.59	11.40	8.30
	40	8.30	11.39	6.47	10.25	8.87	10.59	13.92
	57	6.62	13.11	8.87	9.02	7.80	12.35	10.46
6	13	8.96	10.17	9.27	10.07	10.18	12.63	14.18
	28	10.35	14.73	7.71	9.92	7.03	10.34	9.74
	42	6.87	13.35	7.16	10.16	10.93	10.34	11.12
	44	6.30	12.01	5.91	9.33	8.17	12.13	10.04
	45	11.69	12.85	9.08	9.44	9.54	12.10	17.26
	55	7.44	14.90	5.90	8.10	9.53	12.39	12.14
	61	7.88	11.07	7.76	9.69	10.80	12.15	14.90
	67	5.89	16.22	7.13	9.75	9.23	12.71	11.05

4. Effect of Disease and Treatment on Right Hind Paw Response to Thermal Stimulation (latency; in seconds):

Group	Rat	STUDY DAY						
		0	28	31	38	45	52	59
1	1	11.75	12.44	15.25	11.00	11.24	10.96	14.63
	2	10.96	10.04	11.65	10.98	14.55	11.61	12.32
	3	11.63	11.83	11.19	12.50	11.31	13.43	13.98
	4	8.06	12.04	13.09	13.56	10.95	13.12	11.97
	5	8.69	11.00	10.66	12.14	14.10	11.26	13.97
	6	10.12	12.17	10.80	16.95	12.82	18.57	15.56
	7	8.93	11.23	9.71	11.75	11.35	11.53	13.31
	11	8.18	10.02	11.15	10.14	10.74	11.62	11.39
2	14	8.20	12.03	9.47	7.99	7.64	7.34	8.90
	15	11.25	12.39	8.14	6.50	7.20	8.57	9.52
	33	7.63	12.17	6.91	6.60	8.26	7.92	7.67
	34	15.50	12.21	6.58	6.83	7.95		
	41	8.78	11.35	7.66	9.66	7.24	8.20	9.91
	43	11.45	11.53	5.71	8.79	8.66	11.52	10.08
	48	15.45	13.32	6.43	8.48	5.06	7.50	7.92
	69	6.03	11.02	6.49	6.89	8.02	8.28	11.72
3	22	5.80	13.04	6.52	7.52	7.23	8.66	13.04
	27	9.62	10.81	8.78	8.47	7.72	7.78	10.75
	29	7.76	13.09	7.42	10.87	8.58	10.91	10.11
	31	9.74	14.93	7.76	6.16	7.61	13.47	
	36	10.05	11.14	8.92	10.09	8.73	11.93	11.45
	46	4.58	11.31	8.59	4.69	6.97	5.30	9.65
	47	9.10	15.59	7.58	6.50	5.60	6.63	4.82
	64	6.40	11.85	8.14	5.62	8.50	9.10	6.50
4	19	11.24	10.64	7.72	10.93	8.45	8.33	9.79
	35	10.94	13.10	6.95	9.76	8.41	11.47	10.44
	37	5.93	11.55	7.84	5.77	8.61	8.05	11.34
	54	6.55	8.83	5.34	7.34	8.06	7.66	10.79
	58	8.15	10.58	7.95	7.19	5.79	6.99	8.73
	59	8.63	14.15	7.75	8.41	6.50	8.83	8.46
	60	9.64	9.00	7.22	9.03	8.31	11.30	7.54
	68	9.23	9.18	6.81	5.81	7.26	5.82	8.09

4. Effect of Disease and Treatment on Right Hind Paw Response to Thermal Stimulation (latency; in seconds): continued

Group	Rat	STUDY DAY						
		0	28	31	38	45	52	59
5	8	8.39	4.96	5.20	8.75	8.18	8.26	11.15
	16	10.85	14.15	7.45	8.52	9.03	16.26	13.91
	20	10.39	13.25	9.30	9.09	10.50	9.51	9.73
	23	5.35	13.04	7.79	11.51	8.57	10.04	11.05
	24	6.86	10.13	9.86	6.48	9.66	8.55	11.56
	32	6.54	11.17	7.28	9.58	8.33	10.53	7.83
	40	6.70	12.85	7.91	9.14	9.37	12.11	15.57
	57	6.20	11.09	8.51	8.01	8.80	9.58	9.86
6	13	11.10	11.38	9.02	10.07	10.71	11.12	10.40
	28	7.33	17.59	7.47	10.78	8.33	10.09	11.63
	42	6.18	11.49	7.87	11.26	9.75	12.17	11.63
	44	9.43	10.62	7.44	9.58	6.80	12.61	10.69
	45	8.08	13.91	9.07	10.77	9.31	10.45	11.47
	55	9.10	10.63	4.97	8.50	10.44	11.46	12.07
	61	8.11	14.91	6.89	10.23	9.98	13.73	12.52
	67	6.44	11.25	7.59	7.88	9.94	12.14	12.09

5. Effect of Disease and Treatment on Left Hind Paw Response to Mechanical Stimulation (paw withdrawal pressure; in grams)

Group	Rat	STUDY DAY						
		0	28	31	38	45	52	59
1	1	300.00	26.00	60.0	6.00	26.00	8.00	15.00
	2	300.00	300.00	26.0	10.00	60.00	26.00	10.00
	3	15.00	8.00	8.0	6.00	6.00	6.00	4.00
	4	10.00	8.00	15.0	10.00	6.00	10.00	8.00
	5	100.00	26.00	26.0	15.00	15.00	26.00	6.00
	6	8.00	15.00	60.0	8.00	15.00	8.00	10.00
	7	10.00	60.00	26.0	15.00	8.00	6.00	4.00
	11	100.00	10.00	10.0	8.00	15.00	8.00	8.00
2	14	10.00	1.40	2.00	2.00	0.60	0.60	0.60
	15	10.00	1.00	0.40	0.60	0.40	0.40	0.60
	33	8.00	0.16	0.16	0.40	0.07	0.16	0.16
	34	26.00	4.00	0.60	0.60	1.00		
	41	8.00	10.00	1.40	0.60	1.00	0.40	1.00
	43	26.00	2.00	0.16	1.40	0.60	1.40	1.40
	48	300.00	60.00	4.00	1.00	1.00	0.60	1.40
	69	100.00	0.16	1.00	1.40	1.40	1.40	0.6

5. Effect of Disease and Treatment on Left Hind Paw Response to Mechanical Stimulation (paw withdrawal pressure; in grams) *continued*

Group	Rat	STUDY DAY						
		0	28	31	38	45	52	59
3	22	15.00	1.00	0.60	2.00	0.60	2.00	10.00
	27	8.00	1.40	1.40	0.60	1.00	2.00	10.00
	29	8.00	0.40	0.07	0.07	0.16	0.40	0.60
	31	8.00	0.40	0.60	0.60	1.00	0.40	
	36	10.00	10.00	1.40	2.00	1.40	0.60	4.00
	46	26.00	6.00	0.60	0.40	0.16	0.60	2.00
	47	10.00	10.00	1.40	1.40	1.00	0.40	1.40
	64	100.00	0.40	1.00	4.00	0.40	0.16	2.00
4	19	15.00	8.00	0.60	0.16	1.40	4.00	1.00
	35	10.00	0.60	15.00	0.40	0.16	2.00	2.00
	37	8.00	0.40	0.60	0.40	0.16	1.60	0.60
	54	300.00	0.40	0.16	0.16	1.00	0.60	1.00
	58	60.00	4.00	2.00	2.00	2.00	2.00	1.40
	59	15.00	2.00	0.40	0.16	0.40	0.60	0.40
	60	60.00	26.00	0.40	1.40	2.00	0.40	0.60
	68	26.00	4.00	1.40	2.00	0.16	0.60	1.40
5	8	10.00	10.00	6.00	0.16	4.00	2.00	6.00
	16	300.00	4.00	0.40	1.00	1.00	1.00	2.00
	20	60.00	4.00	1.00	0.40	6.00	0.60	0.60
	23	8.00	6.00	2.00	0.40	0.60	1.00	4.00
	24	15.00	1.00	0.07	0.16	0.60	1.00	0.40
	32	15.00	0.40	0.16	0.40	1.40	2.00	0.60
	40	8.00	0.07	0.07	1.60	4.00	1.00	4.00
	57	15.00	4.00	0.60	4.00	4.00	2.00	4.00
6	13	8.00	2.00	4.00	0.60	1.40	0.60	4.00
	28	8.00	1.00	1.40	0.40	0.60	0.40	1.00
	42	6.00	0.60	1.00	4.00	1.00	8.00	0.60
	44	26.00	1.40	0.60	2.00	4.00	8.00	15.00
	45	26.00	0.16	0.40	0.60	1.40	6.00	15.00
	55	26.00	0.60	0.40	1.40	10.00	8.00	6.00
	61	180.00	1.00	1.00	4.00	8.00	8.00	1.40
	67	8.00	1.00	0.40	4.00	1.00	6.00	4.00

6. Effect of Disease and Treatment on Right Hind Paw Response to Mechanical Stimulation (paw withdrawal pressure; in grams)

Group	Rat	STUDY DAY						
		0	28	31	38	45	52	59
1	1	300.00	8.00	60.00	15.00	60.00	10.00	15.00
	2	300.00	60.00	26.00	26.00	60.00	15.00	15.00
	3	300.00	6.00	8.00	8.00	6.00	6.00	4.00
	4	8.00	8.00	15.00	4.00	6.00	15.00	6.00
	5	100.00	300.00	26.00	4.00	15.00	8.00	8.00
	6	8.00	8.00	60.00	8.00	15.00	10.00	8.00
	7	8.00	8.00	26.00	8.00	8.00	8.00	15.00
	11	15.00	10.00	10.00	8.00	15.00	8.00	10.00
2	14	10.00	2.00	2.00	0.40	1.50	0.60	0.60
	15	26.00	1.40	0.40	0.40	0.40	0.40	0.40
	33	8.00	0.16	0.16	0.40	0.07	0.60	0.16
	34	8.00	1.40	0.60	0.40	1.00		
	41	8.00	6.00	1.40	1.40	1.00	0.40	1.00
	43	100.00	2.00	0.16	0.60	0.60	0.60	0.40
	48	300.00	26.00	1.40	0.40	1.00	0.60	1.40
	69	100.00	0.16	1.00	0.40	1.4	1.40	0.6
3	22	15.00	1.00	0.60	1.40	0.60	1.40	6.00
	27	8.00	1.40	1.40	1.00	1.00	2.00	10.00
	29	8.00	0.40	0.07	0.07	0.16	0.40	0.60
	31	8.00	0.60	0.60	0.40	0.40	0.40	
	36	8.00	15.00	15.00	0.60	1.00	0.40	4.00
	46	26.00	2.00	0.60	0.40	0.40	0.60	2.00
	47	15.00	6.00	2.00	1.40	1.00	1.40	8.00
	64	100.00	0.60	1.00	4.00	0.4	0.16	6.00
4	19	15.00	8.00	0.60	0.16	1.00	4.00	0.40
	35	10.00	0.40	1.40	0.40	0.60	1.40	2.00
	37	8.00	6.00	0.40	0.07	0.16	0.16	1.00
	54	100.00	0.40	0.16	0.07	1.00	0.60	0.60
	58	60.00	6.00	4.00	2.00	2.00	1.00	1.00
	59	10.00	6.00	0.40	0.16	0.40	2.00	0.40
	60	60.00	60.00	1.00	1.00	1.00	2.00	1.00
	68	15.00	4.00	1.40	2.00	0.16	1.00	1.40

6. Effect of Disease and Treatment on Right Hind Paw Response to Mechanical Stimulation (paw withdrawal pressure; in grams) *continued*

Group	Rat	STUDY DAY						
		0	28	31	38	45	52	59
5	8	10.00	6.00	6.00	0.07	4.00	2.00	4.00
	16	300.00	6.00	0.40	1.00	0.40	2.00	1.00
	20	60.00	2.00	1.00	0.40	6.00	1.00	0.60
	23	10.00	2.00	2.00	0.60	0.60	2.00	1.40
	24	15.00	1.00	0.07	0.16	1.40	1.00	0.40
	32	60.00	0.40	0.16	0.07	2.00	0.40	0.60
	40	8.00	0.07	0.07	0.60	4.00	2.00	0.40
	57	15.00	4.00	1.00	8.00	4.00	2.00	4.00
6	13	10.00	4.00	4.00	2.00	4.00	0.60	1.40
	28	10.00	1.00	4.00	0.60	2.00	6.00	1.00
	42	8.00	4.00	2.00	4.00	6.00	8.00	15.00
	44	26.00	0.60	0.60	2.00	4.00	8.00	6.00
	45	26.00	0.16	0.60	0.60	1.00	6.00	10.00
	55	100.00	0.40	0.40	1.40	10.00	15.00	10.00
	61	100.00	0.60	1.00	6.00	8.00	6.00	4.00
	67	8.00	1.00	0.40	4.00	1.4	8.00	1.40

Statistical Analysis:

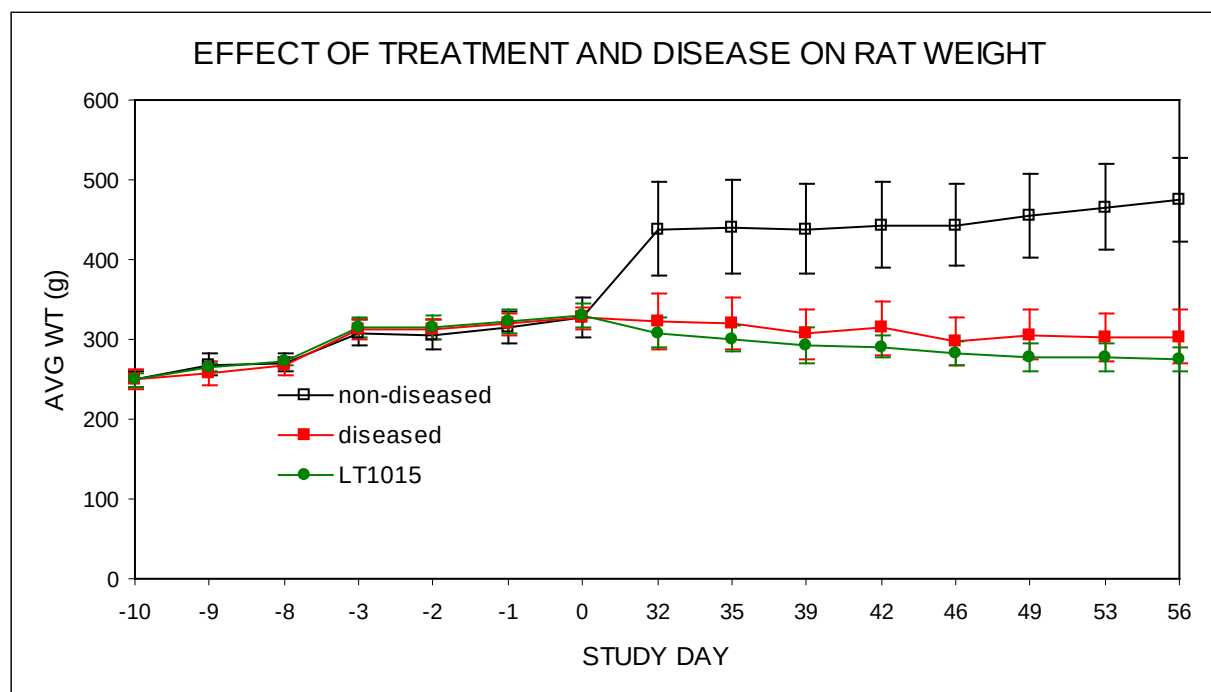
1. Effect of Disease and Treatment on Average Rat Weight (g):

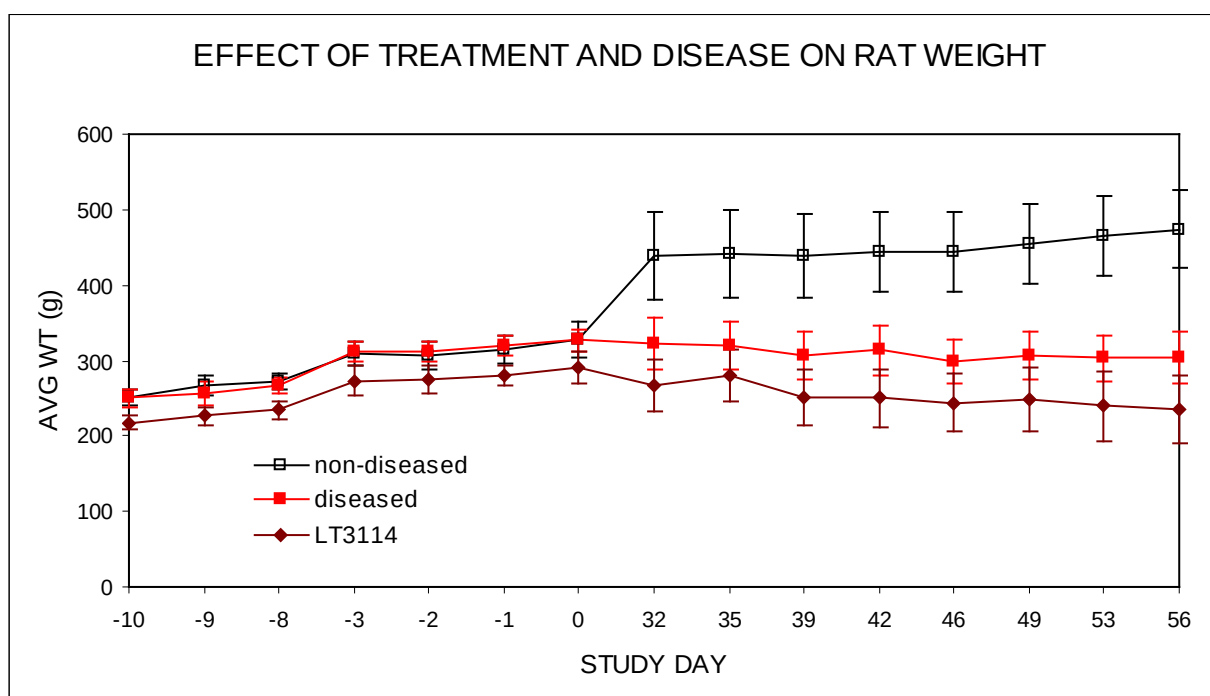
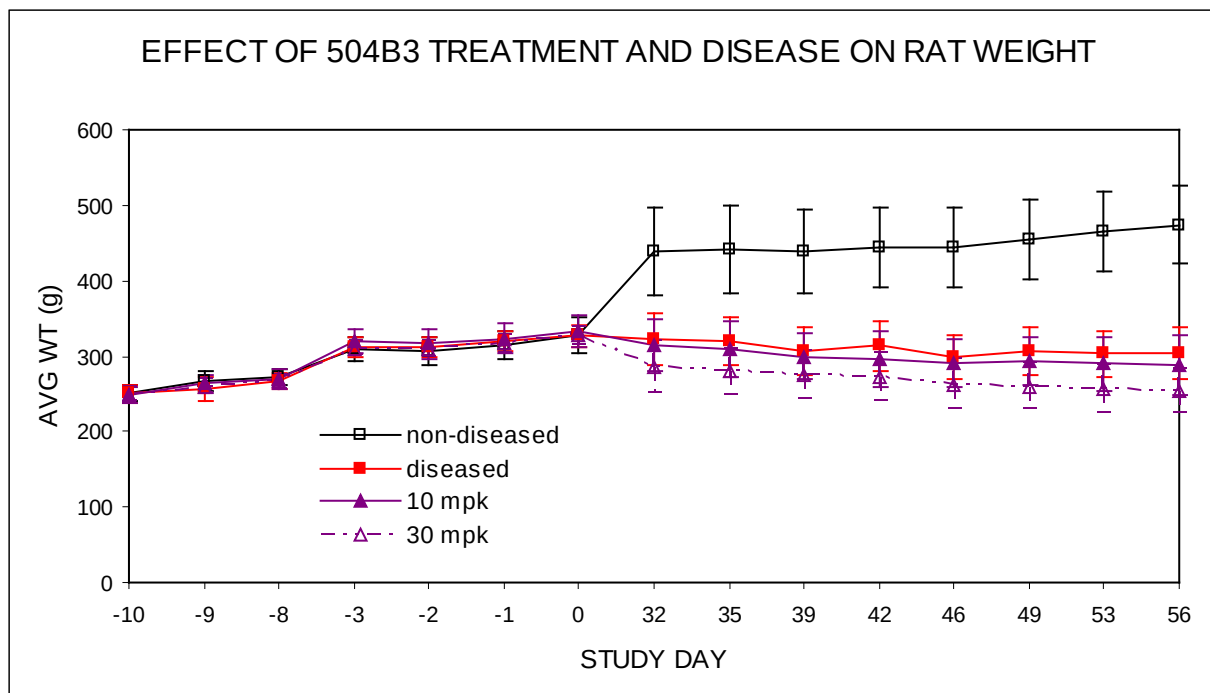
Group	Statistic	STUDY DAY												
		-10	-9	-8	-3	-2	-1	0	32	35	39	42	46	49
1	Mean	250	268	271	309	306	315	328	439	441	439	444	444	455
	SD	10	14	11	15	19	19	24	58	59	56	54	52	53
2	Mean	250	257	267	313	312	319	327	323	320	307	314	298	306
	SD	12	16	11	14	14	13	14	35	31	31	33	30	32
	p vs Group 1	0.97	0.18	0.45	0.61	0.47	0.59	0.93	0.0003	0.0001	4×10^{-5}	4×10^{-5}	2×10^{-5}	2×10^{-5}
3	Mean	249	265	272	314	316	322	331	309	301	292	291	283	277
	SD	10	5	5	13	15	15	15	19	15	22	15	16	18
	p vs Group 2	0.89	0.12	0.15	0.62	0.45	0.49	0.37	0.38	0.19	0.38	0.11	0.30	0.06
4	Mean	250	264	270	320	316	323	333	314	309	299	297	290	293
	SD	11	12	13	15	21	21	20	34	36	30	37	32	32
	p vs Group 2	0.98	0.28	0.49	0.23	0.50	0.50	0.35	0.69	0.63	0.74	0.38	0.76	0.56
5	Mean	250	260	267	311	310	317	327	285	279	274	273	262	260
	SD	9	9	9	10	11	12	11	35	31	31	31	33	30
	p vs Group 2	0.97	0.52	0.86	0.94	0.93	0.87	0.70	0.06	0.02	0.07	0.02	0.06	0.02
6	Mean	218	227	234	273	276	280	290	268	280	251	250	244	248
	SD	10	12	12	20	19	13	21	35	35	37	39	37	41
	p vs Group 2	0.30	0.36	0.33	0.32	0.36	0.33	0.39	0.19	0.15	0.17	0.11	0.21	0.19

Significance (p) was calculated by T-test.

1. Effect of Disease and Treatment on Average Rat Weight (g): *continued*

Group	Statistic	STUDY DAY	
		53	56
1	Mean	466	474
	SD	53	52
2	Mean	303	304
	SD	30	34
	p vs Group 1	7×10^{-6}	6×10^{-6}
3	Mean	278	275
	SD	17	15
	p vs Group 2	0.09	0.07
4	Mean	290	289
	SD	35	39
	p vs Group 2	0.50	0.48
5	Mean	257	255
	SD	31	29
	p vs Group 2	0.01	0.01
6	Mean	240	235
	SD	46	44
	p vs Group 2	0.14	0.10

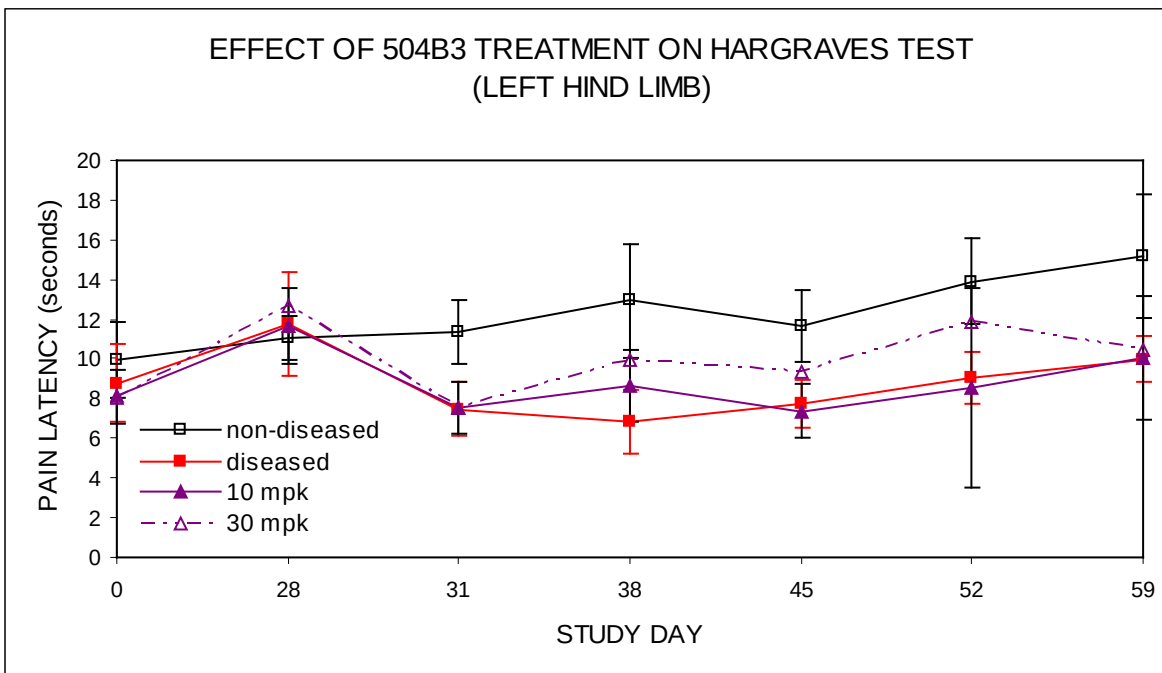
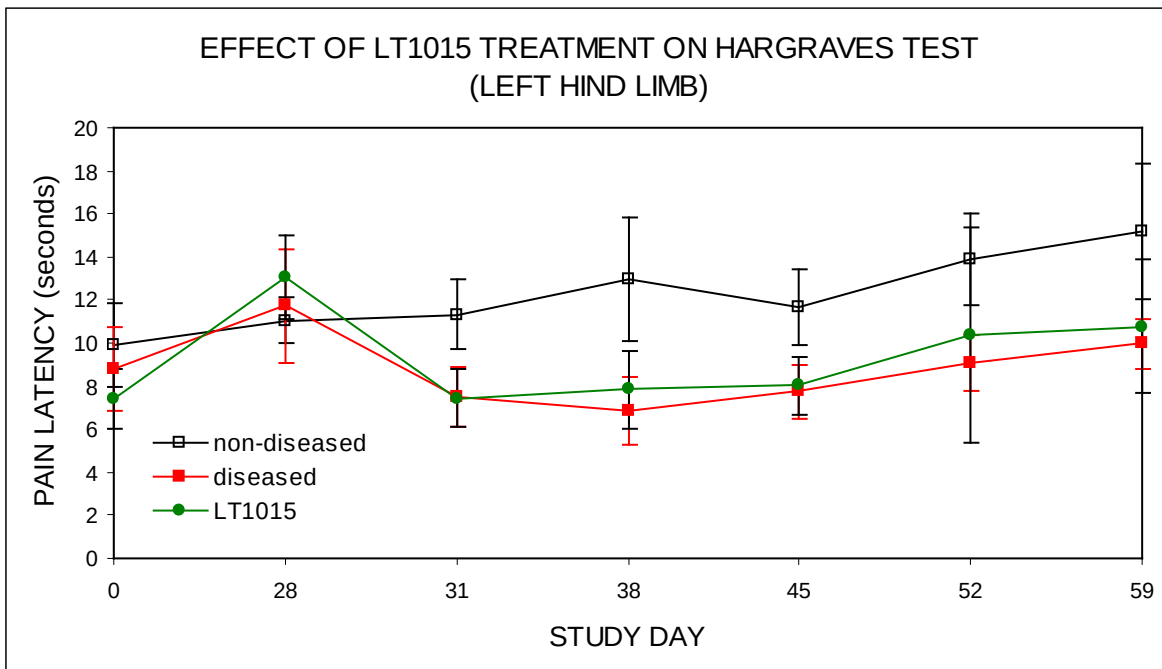


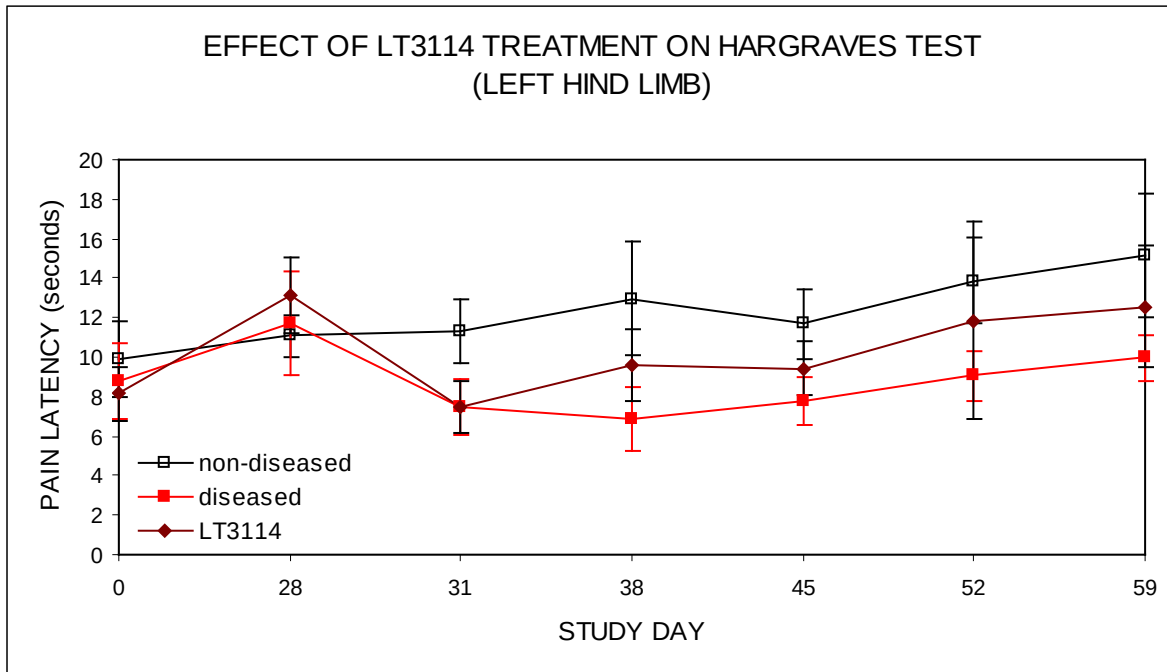


2. Effect of Disease and Treatment on Average Left Hind Paw Response to Thermal Stimulation (latency in seconds):

Group	Statistic	STUDY DAY						
		0	28	31	38	45	52	59
1	Mean	9.90	11.06	11.32	12.94	11.67	13.88	15.15
	SD	1.91	1.08	1.62	2.87	1.79	2.16	3.14
2	Mean	8.79	11.73	7.48	6.85	7.74	9.06	9.99
	SD	1.94	2.62	1.37	1.61	1.23	1.27	1.16
	p vs Group 1	0.27	0.51	0.0002	0.0001	0.0002	0.0002	0.001
3	Mean	7.42	13.04	7.45	7.86	8.02	10.35	10.76
	SD	1.36	1.94	1.31	1.81	1.37	5.00	3.11
	p vs Group 2	0.12	0.27	0.97	0.25	0.67	0.52	0.55
4	Mean	8.12	11.67	7.50	8.66	7.38	8.52	10.06
	SD	1.26	1.55	1.27	1.63	0.93	0.93	1.85
	p vs Group 2	0.43	0.96	0.97	0.04	0.52	0.36	0.93
5	Mean	8.01	12.66	7.50	9.96	9.32	11.86	10.48
	SD	2.69	1.78	1.23	1.28	1.07	3.15	2.09
	p vs Group 2	0.51	0.42	0.97	0.001	0.02	0.05	0.59
6	Mean	8.17	13.16	7.49	9.56	9.42	11.85	12.55
	SD	2.03	2.06	1.26	0.66	1.32	0.96	2.65
	p vs Group 2	0.54	0.24	0.99	0.001	0.02	0.0003	0.03

On DAY 59, the rats in Group 3 were injected subcutaneously with 30 mg/kg gabapentin thirty minutes prior to administration of the Hargreaves test.

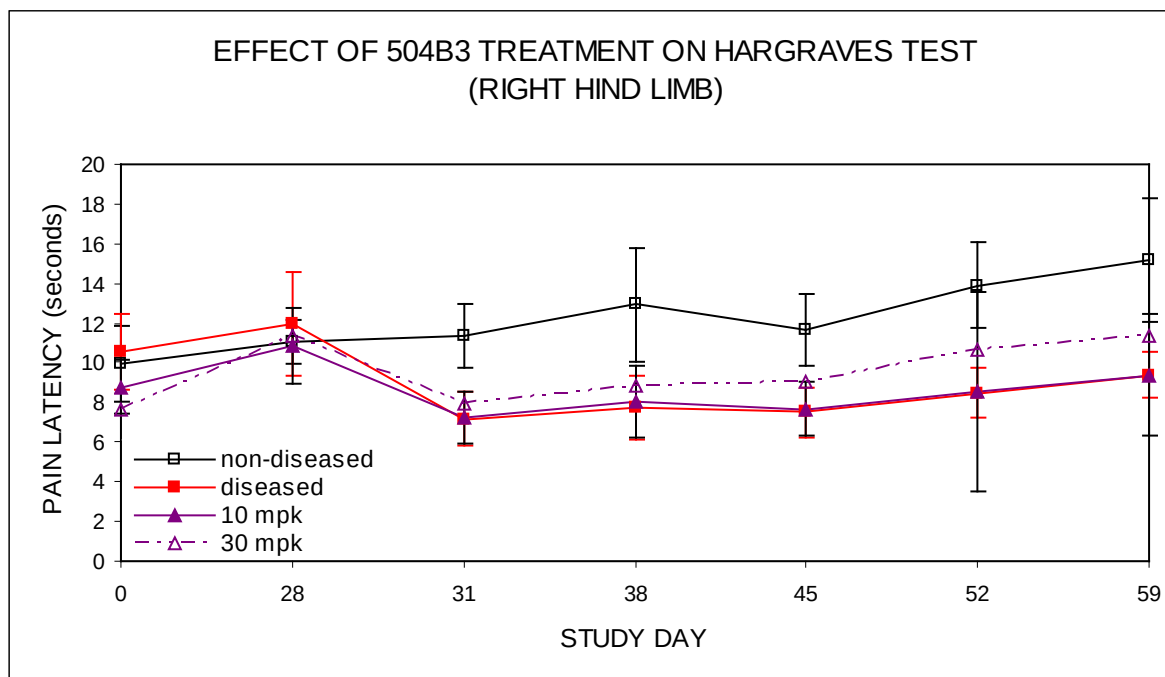
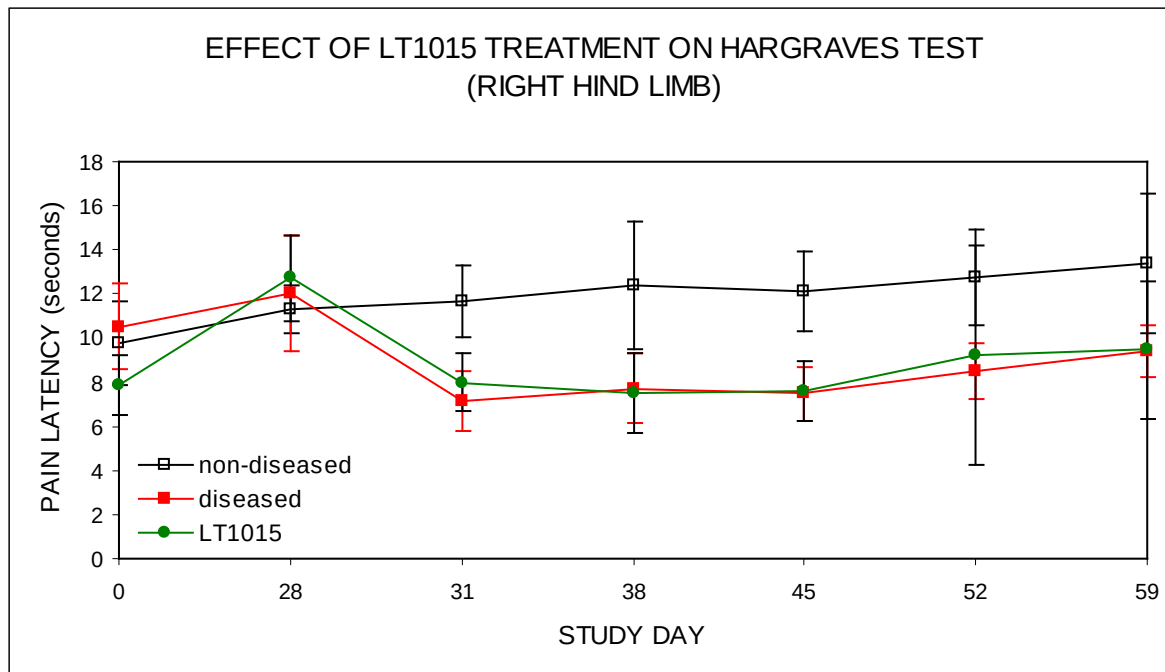


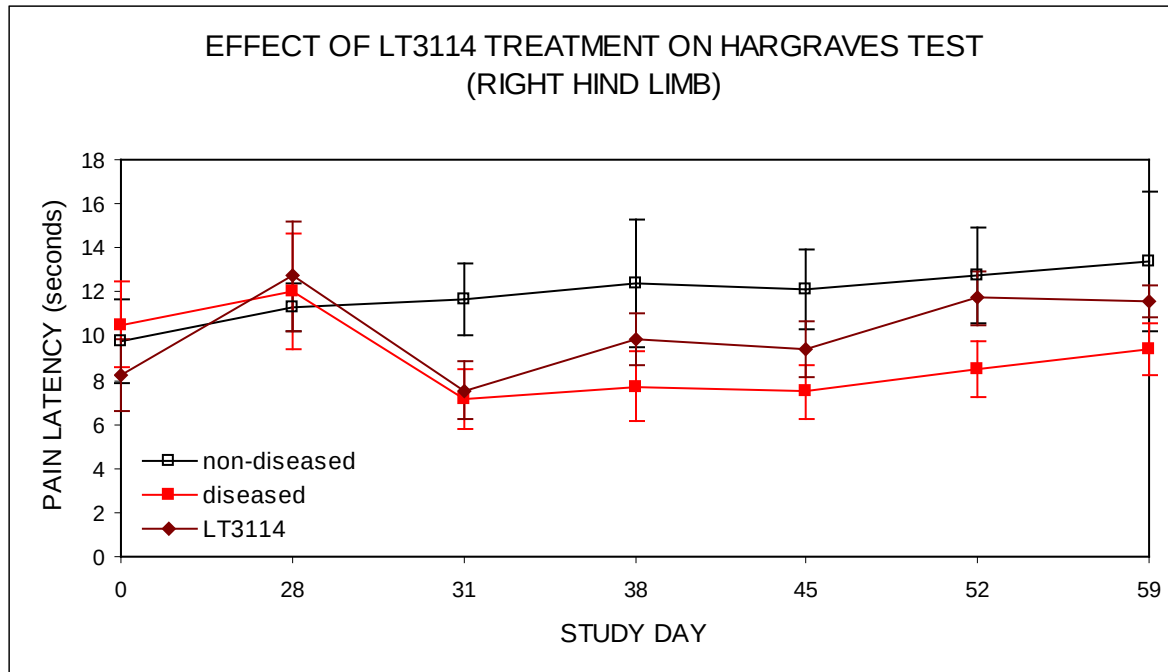


3. Effect of Disease and Treatment on Average Right Hind Paw Response to Thermal Stimulation (latency in seconds):

Group	Statistic	STUDY DAY						
		0	28	31	38	45	52	59
1	Mean	9.79	11.35	11.68	12.38	12.13	12.76	13.39
	SD	1.52	0.94	1.73	2.12	1.49	2.51	1.42
2	Mean	10.54	12.00	7.17	7.72	7.50	8.48	9.39
	SD	3.53	0.72	1.20	1.18	1.10	1.41	1.39
	p vs Group 1	0.59	0.14	3×10^{-5}	9×10^{-5}	6×10^{-6}	0.002	0.0001
3	Mean	7.88	12.72	7.96	7.49	7.62	9.22	9.47
	SD	2.07	1.78	0.81	2.18	1.04	2.75	2.86
	p vs Group 2	0.09	0.31	0.14	0.80	0.83	0.53	0.94
4	Mean	8.79	10.88	7.20	8.03	7.67	8.56	9.40
	SD	1.90	1.96	0.86	1.84	1.05	1.97	1.39
	p vs Group 2	0.24	0.15	0.97	0.70	0.76	0.93	0.99
5	Mean	7.66	11.33	7.91	8.88	9.05	10.61	11.33
	SD	2.02	2.90	1.42	1.42	0.77	2.58	2.44
	p vs Group 2	0.07	0.54	0.28	0.10	0.006	0.07	0.09
6	Mean	8.22	12.72	7.54	9.88	9.41	11.72	11.56
	SD	1.63	2.51	1.29	1.17	1.28	1.19	0.72
	p vs Group 2	0.11	0.45	0.56	0.002	0.007	0.0003	0.002

On DAY 59, the rats in Group 3 were injected subcutaneously with 30 mg/kg gabapentin thirty minutes prior to administration of the Hargreaves test.

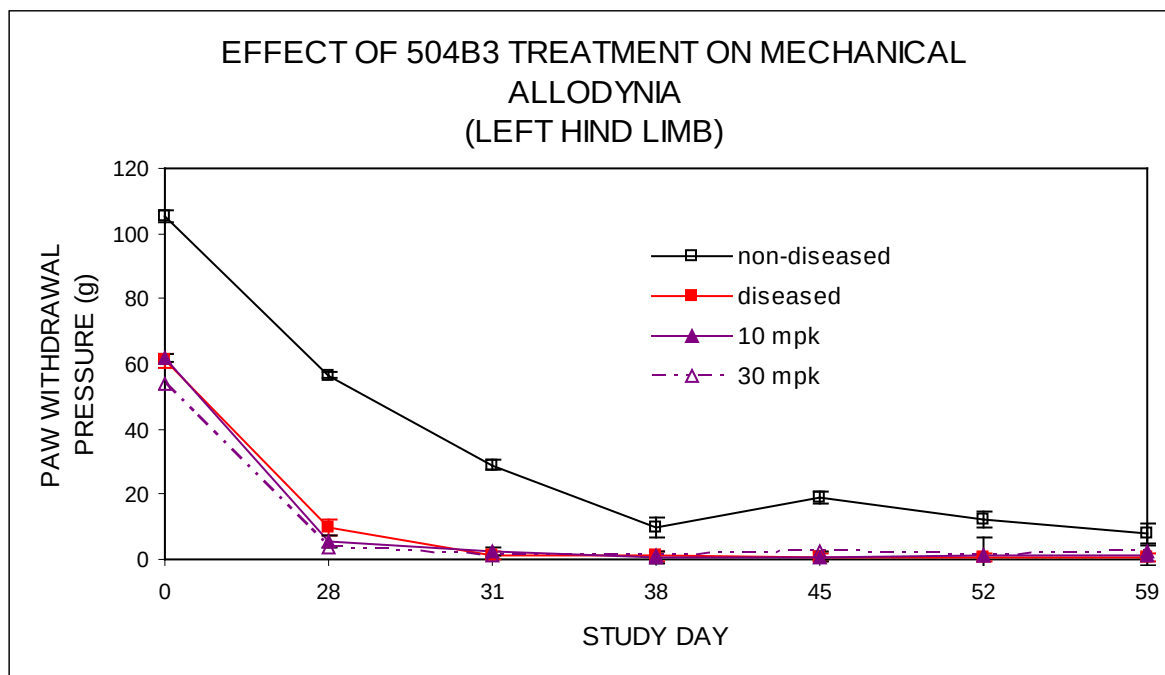
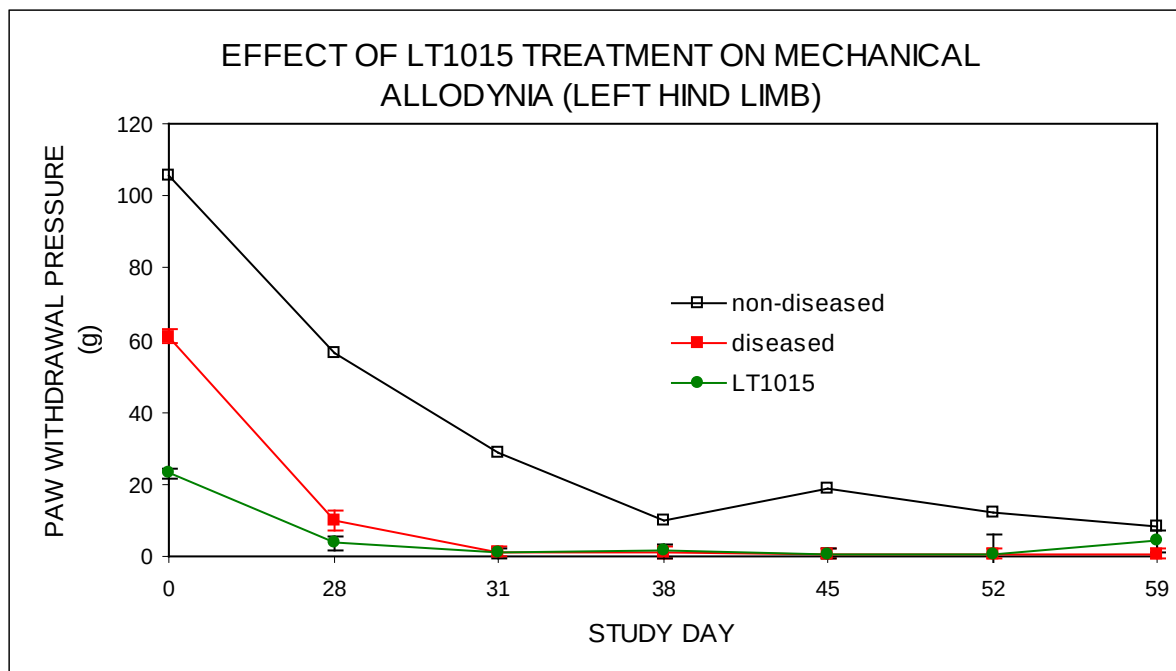


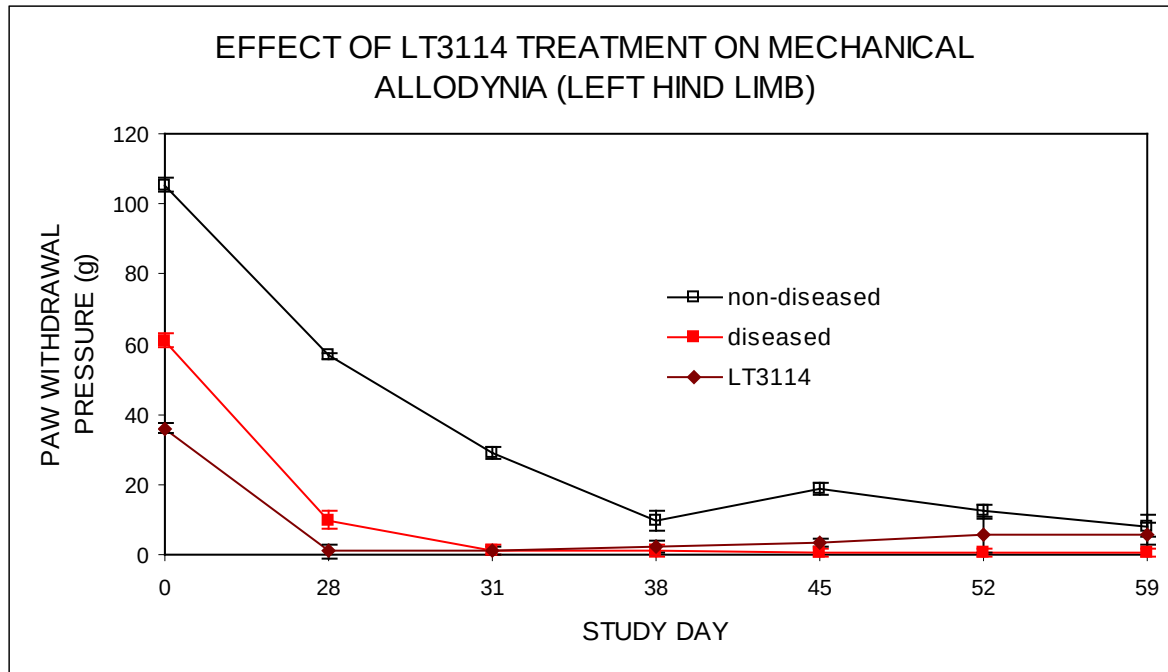


4. Effect of Disease and Treatment on Average Left Hind Paw Response to Mechanical Stimulation (paw withdrawal pressure in g)

Group	Statistic	STUDY DAY						
		0	28	31	38	45	52	59
1	Mean	105.4	56.6	28.9	9.75	18.88	12.25	8.13
	SEM	44.7	35.3	7.3	1.3	6.32	3.03	1.3
2	Mean	61.00	9.84	1.22	1.00	0.76	0.71	0.82
	SEM	35.8	7.3	0.5	0.2	0.1	0.2	0.2
	p vs Group 1	0.45	0.22	0.002	8×10^{-6}	0.01	0.004	0.0002
3	Mean	23.13	3.70	0.88	1.38	0.72	0.82	4.29
	SEM	11.2	1.5	0.2	0.5	0.2	0.3	1.4
	p vs Group 2	0.33	0.42	0.51	0.45	0.84	0.74	0.04
4	Mean	61.75	5.68	2.57	0.84	0.91	1.48	1.05
	SEM	34.8	3.0	1.8	0.3	0.3	0.4	0.2
	p vs Group 2	0.99	0.60	0.48	0.65	0.65	0.15	0.40
5	Mean	53.88	3.68	1.29	1.02	2.70	1.33	2.70
	SEM	35.7	1.2	0.7	0.5	0.7	0.2	0.7
	p vs Group 2	0.89	0.42	0.93	0.98	0.02	0.05	0.04
6	Mean	36.00	0.97	1.15	2.13	3.43	5.63	5.88
	SEM	20.8	0.2	0.4	0.6	1.3	1.2	2.1
	p vs Group 2	0.56	0.24	0.92	0.09	0.06	0.002	0.04

On DAY 59, the rats in Group 3 were injected subcutaneously with 30 mg/kg gabapentin thirty minutes prior to evaluation of mechanical stimulation.

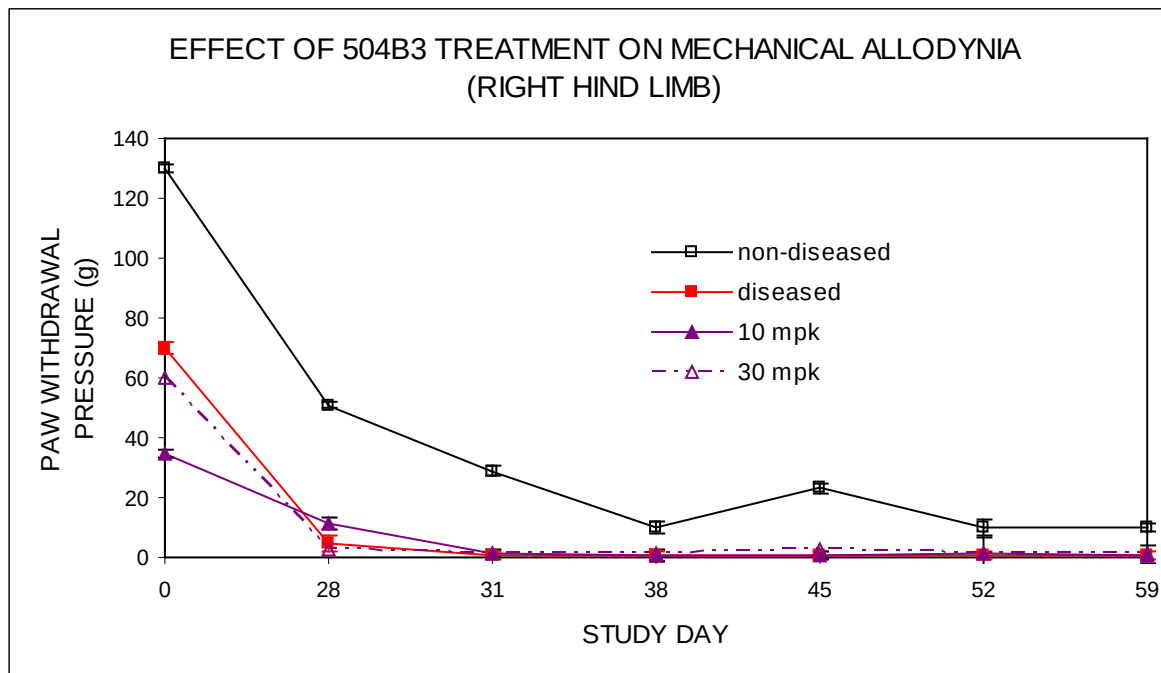
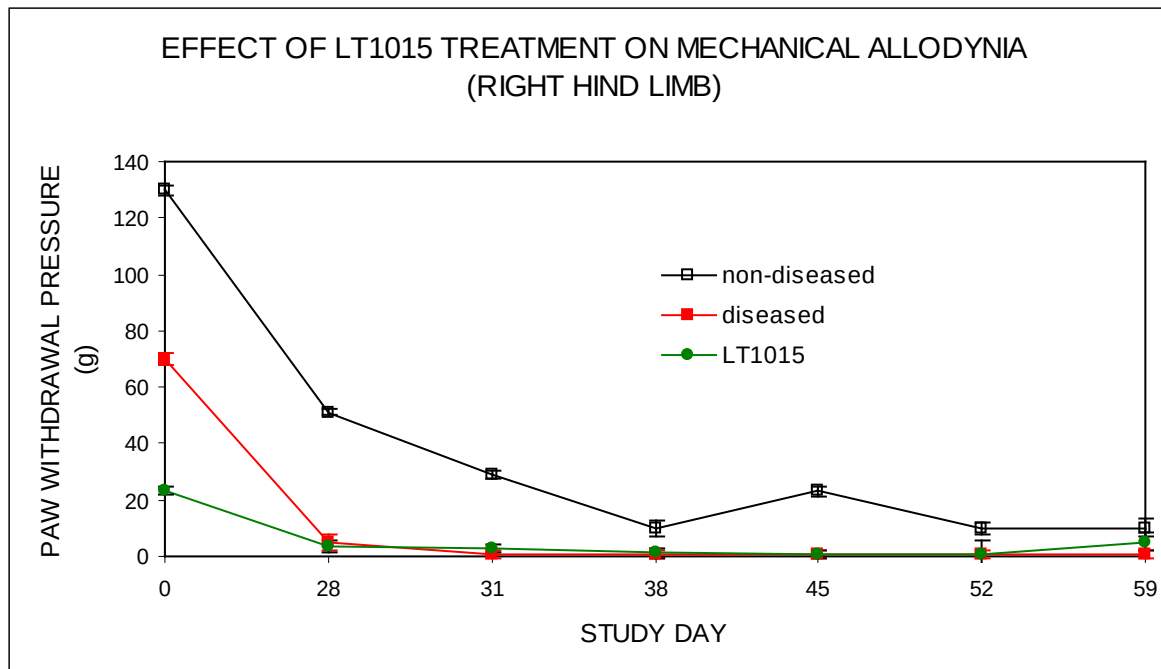


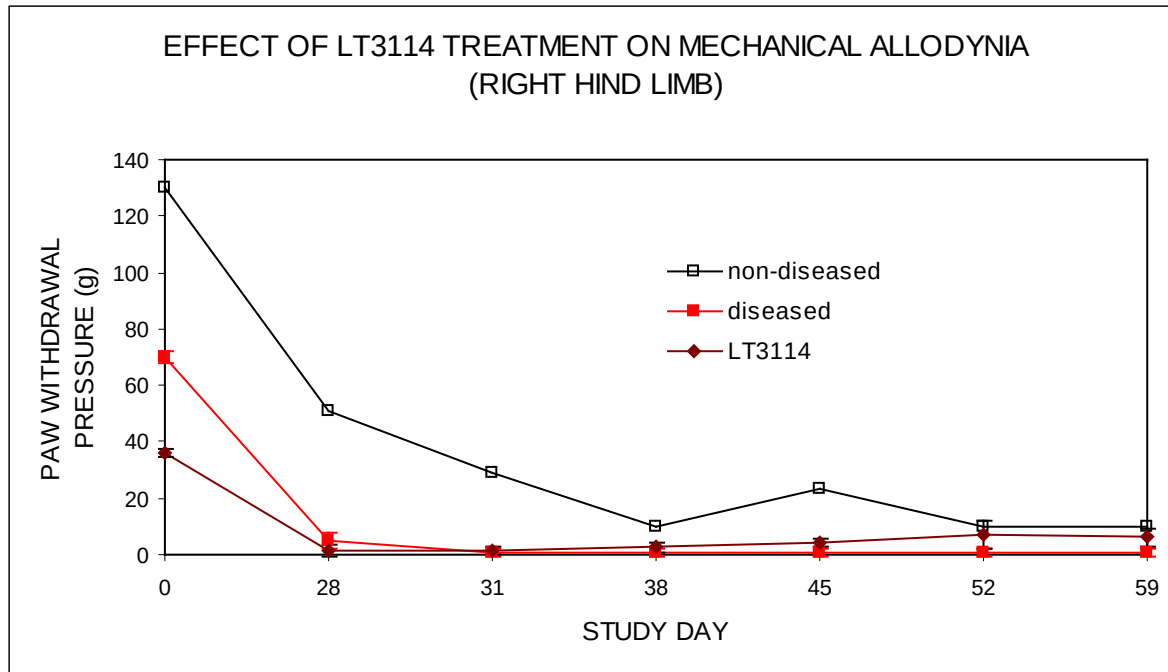


5. Effect of Disease and Treatment on Average Right Hind Paw Response to Mechanical Stimulation (paw withdrawal pressure in g)

Group	Statistic	STUDY DAY						
		0	28	31	38	45	52	59
1	Mean	129.88	51.00	28.88	10.13	23.13	10.00	10.13
	SEM	50.97	36.15	7.25	2.57	8.16	1.18	1.55
2	Mean	70.00	4.89	0.89	0.55	0.87	0.66	0.65
	SEM	35.8	3.1	0.2	0.1	0.2	0.1	0.1
	p vs Group 1	0.35	0.22	0.002	0.002	0.02	6x10 ⁻⁶	0.0001
3	Mean	23.50	3.38	2.66	1.16	0.62	0.85	5.23
	SEM	11.2	1.8	1.8	0.4	0.1	0.2	1.2
	p vs Group 2	0.24	0.68	0.34	0.20	0.25	0.51	0.003
4	Mean	34.75	11.35	1.17	0.73	0.79	1.52	0.98
	SEM	12.1	7.0	0.4	0.3	0.2	0.4	0.2
	p vs Group 2	0.37	0.41	0.58	0.58	0.77	0.09	0.22
5	Mean	59.75	2.68	1.34	1.36	2.80	1.55	1.55
	SEM	35.2	0.8	0.7	1.0	0.7	0.2	0.5
	p vs Group 2	0.84	0.50	0.56	0.41	0.02	0.01	0.16
6	Mean	36.00	1.47	1.63	2.58	4.55	7.20	6.10
	SEM	14.2	0.6	0.5	0.7	1.1	1.4	1.8
	p vs Group 2	0.39	0.29	0.24	0.01	0.01	0.001	0.02

On DAY 59, the rats in Group 3 were injected subcutaneously with 30 mg/kg gabapentin thirty minutes prior to evaluation of mechanical stimulation.



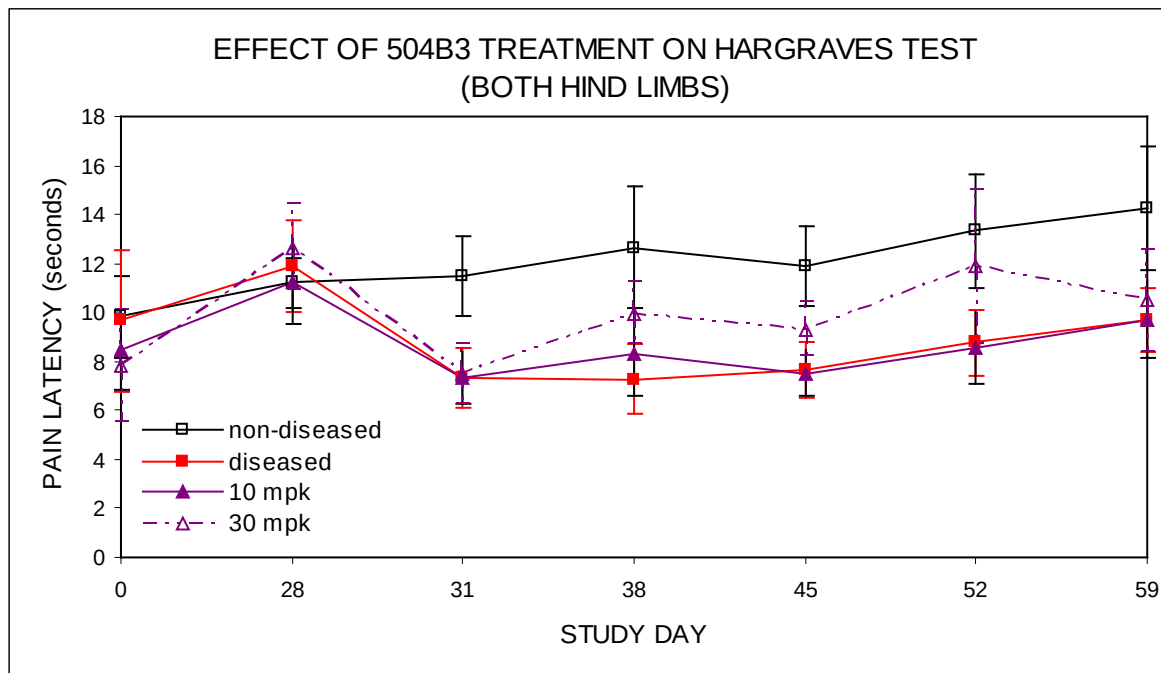
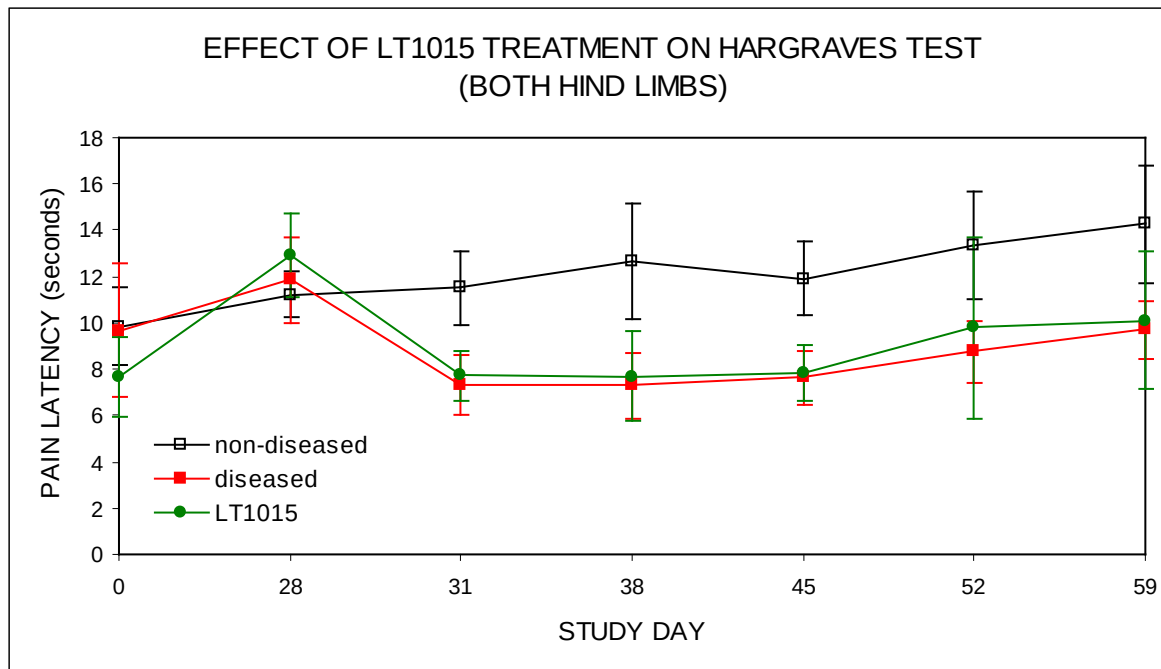


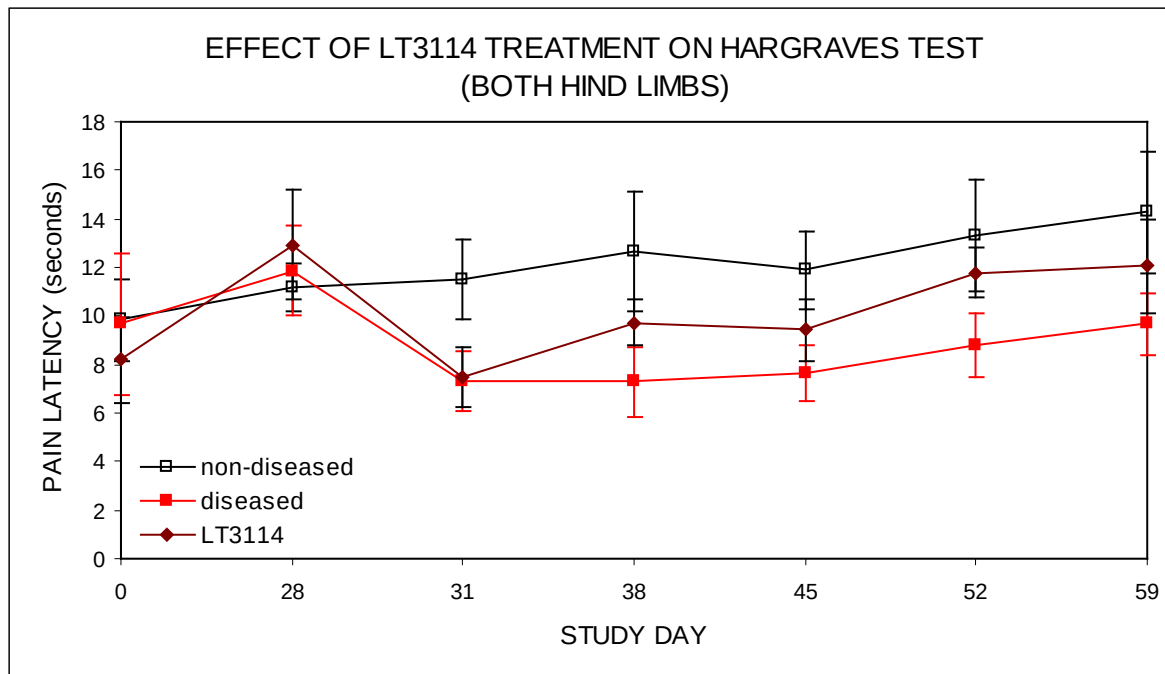
6. Effect of Disease and Treatment on Average Response to Thermal Stimulation
(latency in seconds):

Group	Statistic	STUDY DAY						
		0	28	31	38	45	52	59
1	Mean	9.85	11.21	11.50	12.66	11.90	13.32	14.27
	SD	1.67	0.99	1.63	2.46	1.61	2.33	2.52
2	Mean	9.66	11.87	7.32	7.28	7.62	8.77	9.69
	SD	2.90	1.86	1.25	1.43	1.13	1.33	1.27
	p vs Group 1	0.83	0.22	4×10^{-9}	2×10^{-8}	1×10^{-9}	6×10^{-7}	1×10^{-6}
3	Mean	7.65	12.88	7.71	7.68	7.82	9.79	10.12
	SD	1.71	1.81	1.08	1.94	1.19	3.94	2.95
	p vs Group 2	0.02	0.13	0.36	0.52	0.64	0.37	0.63
4	Mean	8.46	11.28	7.35	8.34	7.53	8.54	9.73
	SD	1.59	1.75	1.06	1.71	0.97	1.49	1.62
	p vs Group 2	0.15	0.36	0.96	0.07	0.80	0.66	0.94
5	Mean	7.83	12.66	7.50	9.96	9.32	11.86	10.48
	SD	2.30	1.78	1.23	1.28	1.07	3.15	2.09
	p vs Group 2	0.06	0.87	0.40	0.0002	0.0002	0.006	0.08
6	Mean	8.20	12.94	7.51	9.72	9.42	11.78	12.06
	SD	1.78	2.23	1.23	0.93	1.25	1.05	1.94
	p vs Group 2	0.09	0.15	0.67	3×10^{-6}	0.0002	1×10^{-7}	0.001

The data is the result of the combined measurements from both the right and left hind limbs.

On DAY 59, the rats in Group 3 were injected subcutaneously with 30 mg/kg gabapentin thirty minutes prior to evaluation of mechanical stimulation.



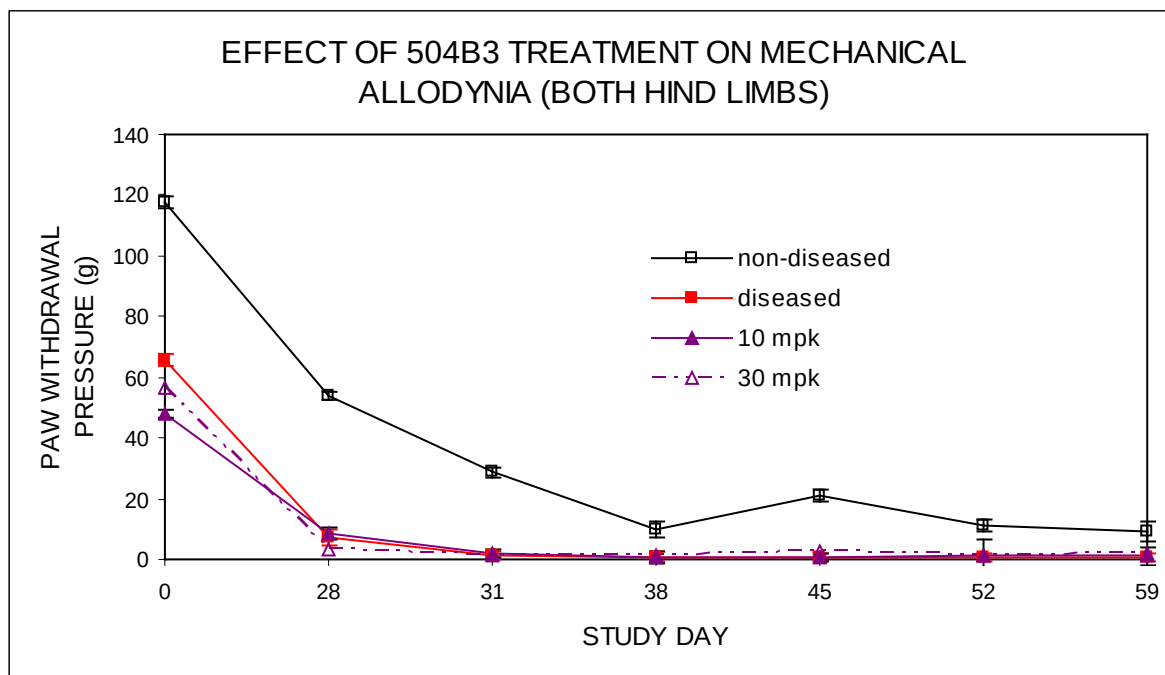
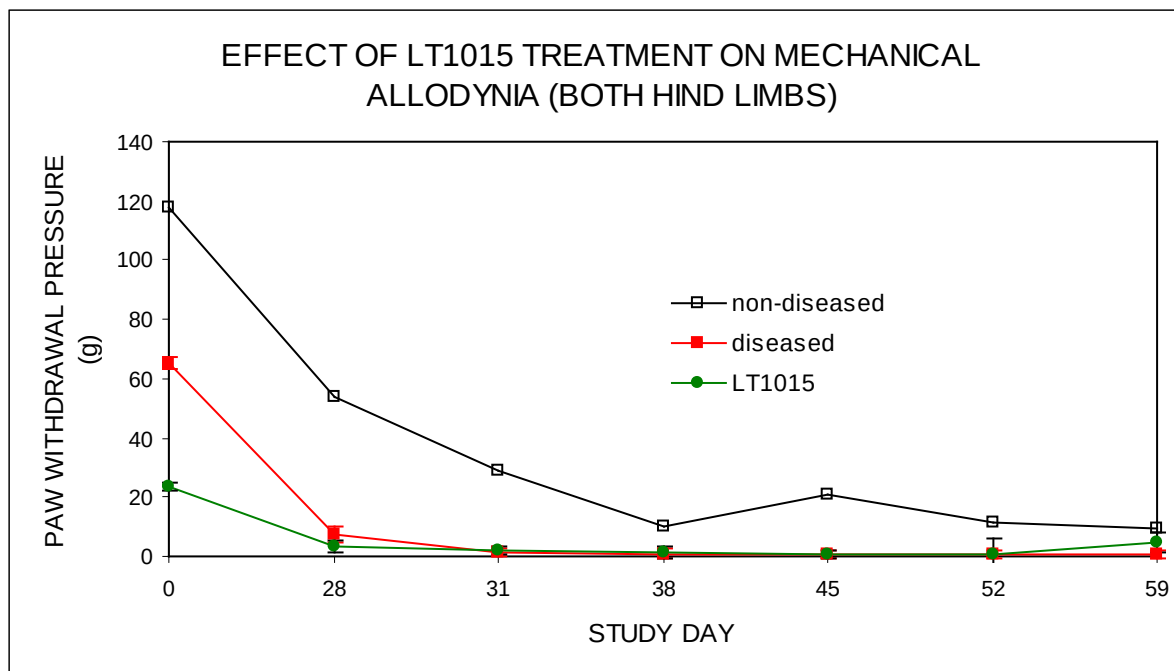


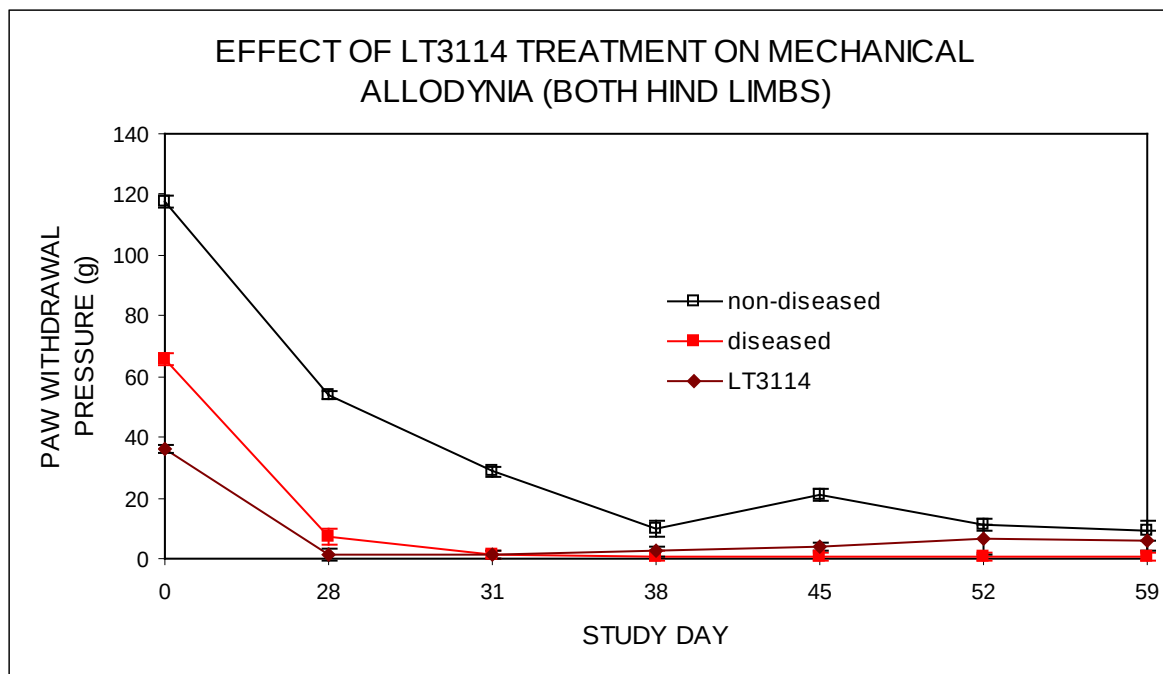
7. Effect of Disease and Treatment on Average Response to Mechanical Stimulation (paw withdrawal pressure in g)

Group	Statistic	STUDY DAY						
		0	28	31	38	45	52	59
1	Mean	117.63	53.81	28.88	9.94	21.00	11.13	9.13
	SEM	32.88	24.42	4.96	1.38	5.02	1.60	1.01
2	Mean	65.50	7.37	1.05	0.78	0.82	0.68	0.74
	SEM	24.5	3.9	0.3	0.1	0.1	0.1	0.1
	p vs Group 1	0.21	0.07	4×10^{-6}	3×10^{-7}	0.0004	1×10^{-6}	2×10^{-8}
3	Mean	23.31	3.54	1.77	1.27	0.67	0.83	4.76
	SEM	7.6	1.1	0.9	0.3	0.1	0.2	0.9
	p vs Group 2	0.11	0.35	0.44	0.15	0.33	0.48	0.0003
4	Mean	48.25	8.51	1.87	0.78	0.85	1.50	1.01
	SEM	18.2	3.8	0.9	0.2	0.2	0.3	0.1
	p vs Group 2	0.58	0.83	0.39	0.97	0.87	0.02	0.13
5	Mean	56.81	3.18	1.31	1.19	2.75	1.44	2.13
	SEM	24.2	0.7	0.5	0.5	0.5	0.2	0.5
	p vs Group 2	0.80	0.30	0.64	0.44	0.001	0.0005	0.01
6	Mean	36.00	1.22	1.39	2.35	3.99	6.41	5.99
	SEM	12.2	0.3	0.3	0.4	0.8	0.9	1.3
	p vs Group 2	0.29	0.12	0.44	0.002	0.001	2×10^{-6}	0.001

The data is the result of the combined measurements from both the right and left hind limbs.

On DAY 59, the rats in Group 3 were injected subcutaneously with 30 mg/kg gabapentin thirty minutes prior to evaluation of mechanical stimulation.





Discussion and Conclusions:

Intravenous injection with 55 mg/kg streptozocin resulted in diabetes in rats as defined by a blood glucose value in excess of 300 mg/dl seven days later. Approximately one month after disease induction, the diabetic rats exhibited peripheral neuropathy as indicated by increased sensitivity to thermal (decreased response time) and mechanical (decrease fiber diameter/pressure) stimulation. The rats were sorted into treatment groups based upon the response to thermal stimulation in the left hind paw. Twice weekly intravenous injection with vehicle or antibodies was initiated five weeks after disease initiation and carried out for four weeks.

Twice weekly injection with 50 mg/kg LT1015 (Group 3) had no effect on diseased rat weight gain. This treatment regimen had no ameliorative effect on the diabetic peripheral neuropathy (DPN). On DAY 59, the rats in Group 3 were injected subcutaneously with 30 mg/kg gabapentin thirty minutes prior to evaluation of the peripheral response to thermal and mechanical stimulation. This treatment resulted in a significant increase in the paw pressure required to elicit a response to the mechanical stimulation, but had no effect on the Hargreaves test, relative to the saline-treated (Group 2) diseased animals.

Twice weekly injection with 504B3 affected the diabetic rat weight at the highest dose (30 mg/kg, Group 5). A significant improvement of DPN was observed at this dose starting at week 7.

The LT3114 treatment regimen (Group 6) had no effect on diabetic rat weight. After the first week (DAY 38) of this treatment regimen, the rats exhibited significantly increased pain latency in response to thermal stimulation. This treatment regimen also

yielded a significant diversion from the saline-treated diseased group (2), relative to mechanical stimulation, after one week of treatment.

APPENDIX 1

Mouse IgG Quantification ELISA

B₃ antibody plasma concentrations were quantified using a direct-binding immunoassay (ELISA) method. Goat-anti mouse IgG (gamma-chain specific) UNLB antibody (Southern Biotech 1030-01, 1mg/mL) was diluted 1:1000 in carbonate buffer (100mM NaHCO₃, 33.6 mM Na₂CO₃, pH 9.5). 384 well plates were coated with 15µL/well of this coating solution and incubated at 4°C overnight. The plates were then washed 4 times with PBS and blocked with 50 µL/well blocking buffer (1%BSA, 0.1% Tween-20 in PBS) for 1 hour at room temperature. Samples and standard were prepared on non-binding plates with enough volume to run in duplicate. The standard was prepared by diluting B₃ or 1015 antibody in blocking buffer (1%BSA, 0.1% Tween-20 in PBS) to the following dilutions: 900 ng/ml, 300 ng/ml, 100 ng/ml, 33.33 ng/ml, 11.11 ng/ml, 3.70 ng/ml, 1.23 ng/ml, 0.41 ng/ml, 0.14 ng/mL, 0.05 ng/mL, 0.02 ng/mL, and 0 ng/mL. The samples were diluted in blocking buffer 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1:7290, 1: 21870, 1: 65610, 1: 196380 and 1: 590490 and 15µL of the standard/samples preparation were loaded to each well and incubated at room temperature for 1 hour. Plates were washed 4 times with PBS and 15 µL of 1: 10,000 blocking buffer diluted goat anti-mouse IgG HRP (Southern Biotech 1030-05) was added to each well and incubated for 1 hour at room temperature. Then, the plates were washed 4 times with PBS and developed using 15 µl/well TMB substrate at 4°C. After 5-20 minutes, the reaction was stopped with 1M H₂SO₄, 15µl/well. The OD was measured at 450 nm. Data was analyzed using Graphpad Prism software. The standard ODs were graphed using a four parameter equation and this was used to calculate the mouse IgG in the samples. Excel software was used to calculate the µg/ml of mouse IgG in each sample by correcting the values for the dilution factor.

Human IgG Quantification ELISA

LT3114 antibody plasma concentrations were quantified a direct-binding immunoassay (ELISA) method as described above with the following modifications:

Goat-anti human IgG (gamma-chain specific) UNLB antibody (Southern Biotech 2040-01, 1mg/mL) was diluted 1:1000 in carbonate buffer to coat the plates

Goat anti-human IgG HRP (Southern Biotech 2040-05) diluted 1: 10,000 in blocking buffer was used as secondary antibody.

RESULTS

Group 3=1015 (30 mg/kg)

Group 4=B3 (10 mg/kg)

Group 5=B3 (30 mg/kg)

Sample	[mAb]	SDev
G3-22	316	40
G3-27	234	1
G3-29	298	5
G3-36	158	
G3-46	172	
G3-47	216	4
G3-64	47	4

Sample	[mAb]	SDev
G4-19	80	2
G4-35	89	11
G4-37	69	12
G4-54	6	1
G4-58	63	1
G4-59	18	2
G4-60	69	3
G4-68	1	0

Sample	[mAb]	SDev
G5-8	10	0
G5-16	219	17
G5-20	170	9
G5-23	38	4
G5-24	468	37
G5-32	214	24
G5-40	5	0
G5-57	189	15

Group 6=3114 (30 mg/kg)

Sample	[mAb]	SDev
G6-13	135	6
G6-28	246	28
G6-42	627	16
G6-44	484	36
G6-45	224	23
G6-55	220	13
G6-61	345	33
G6-67	368	85

DATA LOCATION

Notebook# 248 pages 79-82